Per-Peptidoform Site-Determining-Ion Localization Score

Design proposal — the singleton regime for multi-peptidoform reporting

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1. Why we need this

We want to report **more than one peptidoform per peptide** (multiple positional isomers can be genuinely co-present), not just one max-weight winner per isomer group. The data already supports it: each isomer is its own `precursor_idx`, deconvolved per-scan and FDR'd independently, and `precursors_long` carries each peptidoform's best-PSM `iso_weight_fraction_at_scan`.

But the last experiment (10k-amol, 3 reps, 54,066 target-target vs 40,814 target-decoy peptidoforms after the `qval` + `global_qval` gate) showed a hard wall:

population	frac >= 0.75	frac >= 0.90	frac >= 0.95	floor
all peptidoforms	11.5%	10.7%	10.8%	**~10.7%, never < 10%**
contested (>=2 competing siblings)	11.3%	8.2%	6.6%	~6%

The `iso_weight_fraction` cutoff cannot push per-peptidoform FLR below ~10%. The cause is diagnostic: at frac >= 0.75, ~half the confident calls are **singletons** — the peptidoform is alone in its isomer group at its apex scan (`iso_group_size_at_scan == 1`), so `frac = 1.0` *automatically* and carries **zero localization information**. Their target-decoy FLR (~11.6%) sets the floor.

A peptidoform is a singleton at its apex exactly when its sibling isomers elute at a **different RT** — which is the "genuinely co-present isomers" case we want to report. The sibling fraction measures **co-elution competition**; RT-separated isomers don't compete, so the fraction is blind to them. A peptidoform decoy that separates in RT with a clean (permuted) spectrum is indistinguishable from a real second form *by the fraction alone.*

Conclusion: the fraction is the right confidence for the co-eluting (contested) regime, but the RT-resolved (singleton) regime needs a different, spectral score. This document proposes that score.

2. Core idea

For a singleton peptidoform, we can't use co-elution competition, but we *can* ask the spectrum directly: **do the fragment ions that actually distinguish the assigned site from every alternative site support the assigned placement?** Those ions are the **site-determining**

ions. The score is a per-peptidoform, intrinsic measure — it needs only the observed spectrum at the peptidoform's apex scan and the library fragments; it does **not** require the competing isomer to be detected.

Crucially, this is exactly the evidence that ID-level FDR does **not** protect. A loc-decoy passes `qval` mostly on **shared** ions (b/y ions whose cleavage does not span the moved site are identical between real and decoy). It should fail on the **site-determining** ions, because its mod mass sits on the wrong fragments. The site-determining score isolates precisely that discriminating evidence.

3. Definitions

3.1 Candidate sites

For a peptidoform of length L with a variable mod assigned to position a, the **candidate alternatives** are the other acceptor residues (for phospho: the other S/T/Y positions). Localization means: the mod is on a, not on any alternative a'.

3.2 Site-determining ions

A fragment ion **distinguishes** placement a from a' iff its cleavage falls **between** them — the ion contains exactly one of $\{a, a'\}$, so the mod mass (+79.966 for phospho) shifts the ion's m/z for one placement but not the other.

- b_i (prefix, cleavage after residue i) contains positions $1..i$. It is site-determining for $\{a, a'\}$ iff exactly one of $a, a' \le i$.
- y_j (suffix) contains positions $(L - j + 1)..L$. Site-determining iff exactly one of a, a' is in that range.

Worked example. Peptide `ABCSDESFGK` ($L = 10$), phospho candidate sites S_4 and S_7.

```
position :  1  2  3  4  5  6  7  8  9 10
residue  :  A  B  C  S  D  E  S  F  G  K
                     ^cand a=4      ^cand a'=7
b-ions   :  b1 b2 b3 b4 b5 b6 b7 b8 b9
y-ions   :     y9 y8 y7 y6 y5 y4 y3 y2 y1   (y_j covers L-j+1..L)
```

Ions whose cleavage lies in the gap $[4, 7)$ are site-determining:

- b_4, b_5, b_6 — contain S_4 but not S_7 → carry +80 if the site is S_4, unmodified if S_7.
- y_4, y_5, y_6 — contain S_7 but not S_4 → carry +80 if the site is S_7, unmodified if S_4.

So the real peptide (site S_4) shows $b_{4,5,6}$ at +80 and $y_{4,5,6}$ unmodified. A decoy claiming S_7 predicts the opposite and will **not** match the observed peaks on those six ions. The gap of distance $\delta = 3$ yields $\sim 2\delta = 6$ site-determining ions — matching the retention design already in `site_determining.jl`.

3.3 Phospho-specific caveat: neutral loss

pS/pT spectra are dominated by the **neutral loss of H3PO4** (−97.977) from the precursor and from mod-bearing fragments. Neutral-loss peaks are *not* site-determining (the loss can occur from any placement) and, worse, a b_i^{-98} can be isobaric with an unmodified b_i. **The scorer must**

exclude neutral-loss channels from the site-determining set (or model them explicitly), or decoys will look better than they are. This is the single biggest correctness risk (see §9).

4. Score formulations

Let $I(\cdot)$ be the observed intensity of a matched peak (0 if unmatched within mass tolerance), computed at the peptidoform's apex scan.

(A) SDI count — number of matched site-determining ions supporting the assigned site. Simple, but not abundance-normalized and ignores the alternatives' evidence.

(B) SDI intensity fraction —

$$S_B = \frac{\sum_{\text{ion}\in\text{SDI, supports } a} I(\text{ion})}{\sum_{\text{ion}\in\text{SDI}} I(\text{ion at either placement})}$$

Normalizes for abundance; still pools all alternatives.

(C) Competitive site-determining ratio (recommended). Mirror the `iso_weight_fraction` logic but from spectral evidence. For each alternative a':

$$s(a,a') = \frac{\text{support}(a)}{\text{support}(a)+\text{support}(a')}, \quad \text{support}(x) = \sum_{\text{ion}\in\text{SDI}(a,a')} I(\text{ion at placement } x)$$

The peptidoform score is the **worst case over all alternatives**:

$$S_C = \min_{a'\neq a} s(a,a')$$

i.e. "the assigned site must beat *every* alternative on its own discriminating ions." $S_C \to 1$ when the site-determining ions cleanly support a; $S_C \to 0.5$ when ambiguous; $S_C < 0.5$ when they actually favor an alternative.

Undefined case. If some alternative a' has **no observed site-determining ions** ($\text{support}(a) + \text{support}(a') = 0$), the site is **unlocalizable against** a' from this spectrum. This is a real, honest outcome (matches the literature: not every site is localizable). The scorer should return a distinct `n_sdi = 0` flag so we can *report the peptide but not the site*, rather than silently scoring it 0 or 1.

Recommendation: compute all three but calibrate on **(C)**, reporting **(A)** `n_sdi` alongside as the "is this even localizable" guard.

5. Two-regime localization confidence

The per-peptidoform localization decision becomes:

```
if iso_group_size_at_scan >= 2:      # contested / co-eluting
    confidence = iso_weight_fraction_at_scan      # existing; ~6.6% FLR at 0.95
else:                                # singleton / RT-resolved
    if n_sdi == 0:  report peptide, flag site "unlocalizable"
    else:           confidence = S_C  (site-determining ratio)
```

Both confidences are then thresholded to a **single target FLR**, calibrated jointly against the target-decoy population (§6). Longer term, (C) can *replace* the fraction everywhere (it is defined for contested peptidoforms too), but the two-regime split is the minimal first step and lets us keep the already-validated fraction where it works.

6. Calibration & FLR (unchanged framework)

FLR is estimated exactly as now — **target-target vs target-decoy among sequence-targets only, sequence decoys excluded** (they only set the upstream `qval`/`global_qval`). Every passing peptidoform, real or loc-decoy, gets the same S_C computed the same way:

$$\mathrm{FLR}(\tau) = \frac{\#\{\text{target-decoy singletons} : S_C \geq \tau\}}{\#\{\text{target-target singletons} : S_C \geq \tau\}}$$

If S_C is a good localization discriminator, target-decoys — whose assigned site is wrong, so their site-determining ions won't match — collapse to low S_C, and FLR should fall well below the fraction's ~11.6% singleton floor. **The central empirical question this prototype answers: how far?**

7. What we can compute now vs. what needs extraction

Input	Source	Status
apex scan index per peptidoform	`new_best_scan` in `precursors_long`	[have] present
passing set (qval, global_qval)	`precursors_long`	[have] present
target-target / target-decoy label	`is_loc_decoy` join from library	[have] present
candidate sites, b/y positions	`site_determining.jl` (`acceptor_positions`)	[have] built
library fragment m/z + ion type + position + mod	library `fragments` table	[have] present
observed peaks (m/z, intensity) at apex scan	the run's Arrow MS2 spectra	[build] needs a read
fragment<->peak matching within tol	new targeted step	[build] to build

So this is **not** a pure `precursors_long` analysis — it is a **targeted extraction**: for each passing singleton, load its apex scan's peaks from the run's Arrow file (`new_best_scan`), match the library fragments, classify site-determining, compute S_C. It reuses the existing `site_determining.jl` machinery and runs offline on the current 10k results — **no re-search, no library rebuild**. Estimated scope: one focused extraction script (~150–250 lines) + reuse.

8. Validation plan

1. **Decoy calibration curve** — FLR(τ) vs τ on 10k singletons; compare to the ~11.6% fraction floor. Success = a threshold exists with FLR <= 5% at usable coverage.
2. **Ground-truth check on standards** — the 225 Coon standards have known single sites. For the singleton subset of detected standards, compare S_C-thresholded **empirical** FLR (site vs truth) to the **decoy-estimated** FLR. Tests whether the decoy estimate is honest for singletons.

3. **Coverage cost** — IDs retained vs the fraction-only and winner-take-one baselines, at matched FLR.
4. **Unlocalizable rate** — fraction of singletons with `n_sdi = 0` (report-peptide-not-site); this is a real property of the data, not a failure.

9. Risks & open questions

- **Neutral loss (biggest).** Must exclude/model H3PO4 loss channels or decoys look too good. §3.3.
- **Shadow decoys may be too easy — again.** Decoy sites are non-acceptor "shadow" residues; their site-determining ions may be trivially discriminating, so S_C could **underestimate** FLR just as the fraction did in the winner-take-one comparison. The **short-distance-biased** decoy variant is the proper calibrant; SDI scoring and better decoys are complementary, not alternatives.
- **Sparse site-determining ions.** Sites near a terminus, or with the discriminating ions below detection, give small `n_sdi`. Expect a non-trivial unlocalizable fraction — this is honest, but it caps how many singletons we can confidently report.
- **Fragment charge / isotopes / co-isolation.** Match must respect the fragment charge states in the library and avoid crediting a co-isolated precursor's peak. Use the same tolerance the search used.
- D/R **normalization** (carried over): at ~1:1 target-decoy:target-target, D/R is the estimate; revisit vs $D/(R+D)$ if we change the decoy multiplicity.
- **Does S_C subsume the fraction?** Possibly — (C) is defined for contested peptidoforms too. If it matches or beats the fraction there, we unify on one score. To be tested, not assumed.

10. Proposed phased plan

- **Phase 0 (cheap, optional first look).** From `precursors_long` alone: quantify the contested-vs-singleton split, how many peptides have >=2 RT-separated real forms, and the target-decoy rate among singletons vs contested. Sizes the prize before building the extractor. *(~1 script.)*
- **Phase 1 (this proposal).** Build the targeted SDI extractor; compute S_C + `n_sdi` for all passing singletons at 10k; produce the decoy calibration curve (§8.1) and the standards ground-truth check (§8.2). **Decision gate:** does a usable threshold reach <=5% FLR?
- **Phase 2 (if Phase 1 passes).** Wire the two-regime confidence (§5) into the reporting path; sweep to a target FLR; report multi-peptidoform IDs with per-site confidence and an `unlocalizable` flag.
- **Phase 3 (optional).** Replace the fraction with S_C everywhere if it dominates; and/or fold S_C, `n_sdi`, the fraction, and PSM score into a single trained localization model calibrated on the loc-decoys.

Companion docs: `site_determining_fragment_retention.md` (the ion-retention machinery this reuses), `FINDINGS.md` (the fraction-floor result motivating this), `isomer_decoy_shadow_sites.md` (the target-decoy construction).